



1. Description

1.1. Project

Project Name	tcp_server_socket
Board Name	STM32F746G-DISCO
Generated with:	STM32CubeMX 6.10.0
Date	08/05/2025

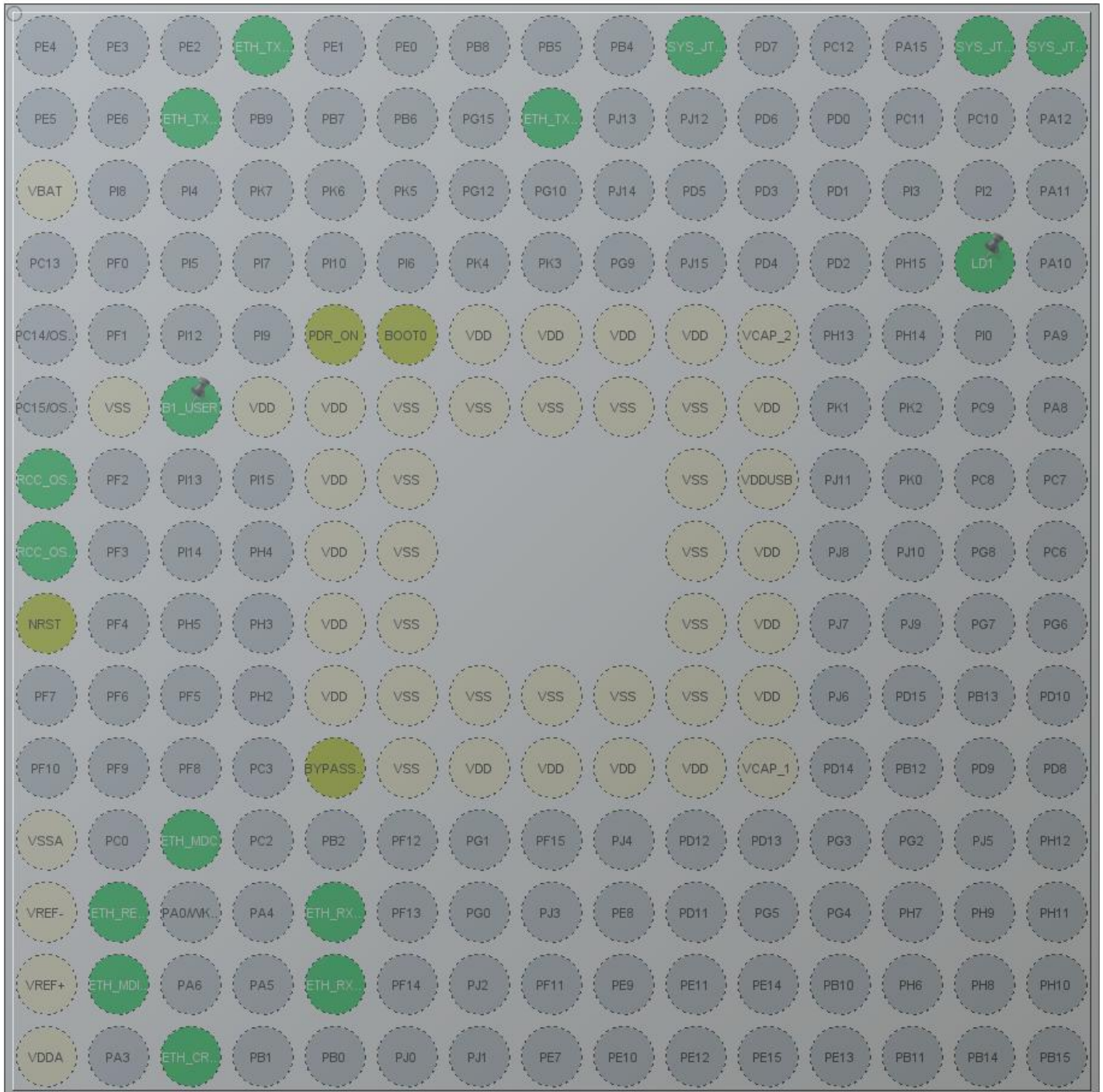
1.2. MCU

MCU Series	STM32F7
MCU Line	STM32F7x6
MCU name	STM32F746NGHx
MCU Package	TFBGA216
MCU Pin number	216

1.3. Core(s) information

Core(s)	Arm Cortex-M7
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2. Pinout Configuration



TFBGA216 (Top view)

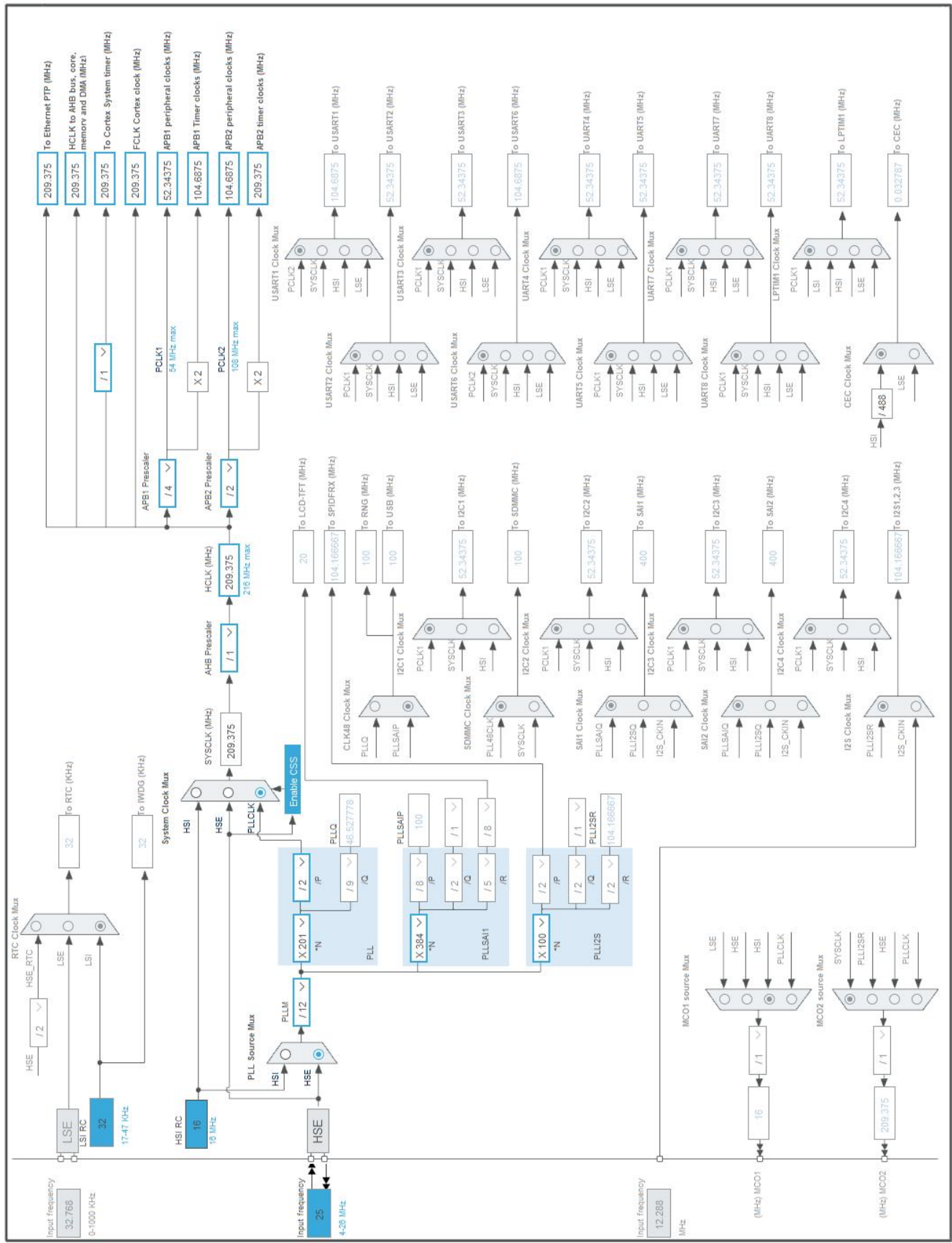
3. Pins Configuration

Pin Number TFBGA216	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A4	PG14	I/O	ETH_TXD1	
A10	PB3	I/O	SYS_JTDO-SWO	
A14	PA14	I/O	SYS_JTCK-SWCLK	
A15	PA13	I/O	SYS_JTMS-SWDIO	
B3	PG13	I/O	ETH_TXD0	
B8	PG11	I/O	ETH_TX_EN	
C1	VBAT	Power		
D14	PI1 *	I/O	GPIO_Output	LD1
E5	PDR_ON	Reset		
E6	BOOT0	Boot		
E7	VDD	Power		
E8	VDD	Power		
E9	VDD	Power		
E10	VDD	Power		
E11	VCAP_2	Power		
F2	VSS	Power		
F3	PI11 *	I/O	GPIO_Input	B1_USER
F4	VDD	Power		
F5	VDD	Power		
F6	VSS	Power		
F7	VSS	Power		
F8	VSS	Power		
F9	VSS	Power		
F10	VSS	Power		
F11	VDD	Power		
G1	PH0/OSC_IN	I/O	RCC_OSC_IN	
G5	VDD	Power		
G6	VSS	Power		
G10	VSS	Power		
G11	VDDUSB	Power		
H1	PH1/OSC_OUT	I/O	RCC_OSC_OUT	
H5	VDD	Power		
H6	VSS	Power		
H10	VSS	Power		
H11	VDD	Power		
J1	NRST	Reset		

Pin Number TFBGA216	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
J5	VDD	Power		
J6	VSS	Power		
J10	VSS	Power		
J11	VDD	Power		
K5	VDD	Power		
K6	VSS	Power		
K7	VSS	Power		
K8	VSS	Power		
K9	VSS	Power		
K10	VSS	Power		
K11	VDD	Power		
L5	BYPASS_REG	Reset		
L6	VSS	Power		
L7	VDD	Power		
L8	VDD	Power		
L9	VDD	Power		
L10	VDD	Power		
L11	VCAP_1	Power		
M1	VSSA	Power		
M3	PC1	I/O	ETH_MDC	
N1	VREF-	Power		
N2	PA1	I/O	ETH_REF_CLK	
N5	PC4	I/O	ETH_RXD0	
P1	VREF+	Power		
P2	PA2	I/O	ETH_MDIO	
P5	PC5	I/O	ETH_RXD1	
R1	VDDA	Power		
R3	PA7	I/O	ETH_CRS_DV	

* The pin is affected with an I/O function

4. Clock Tree Configuration



5. Software Project

5.1. Project Settings

Name	Value
Project Name	tcp_server_socket
Project Folder	D:\OneDrive\Documentos\Os_meus_sistemas_embedidos\STM32F746\tcp_serv
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_F7 V1.17.2
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x2000
Minimum Stack Size	0x4000

5.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

5.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_LWIP_Init	LWIP
4	MX_MBEDTLS_Init	MBEDTLS

1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F7
Line	STM32F7x6
MCU	STM32F746NGHx
Datasheet	DS10916 Rev4

1.2. Parameter Selection

Temperature	25
Vdd	3.3

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

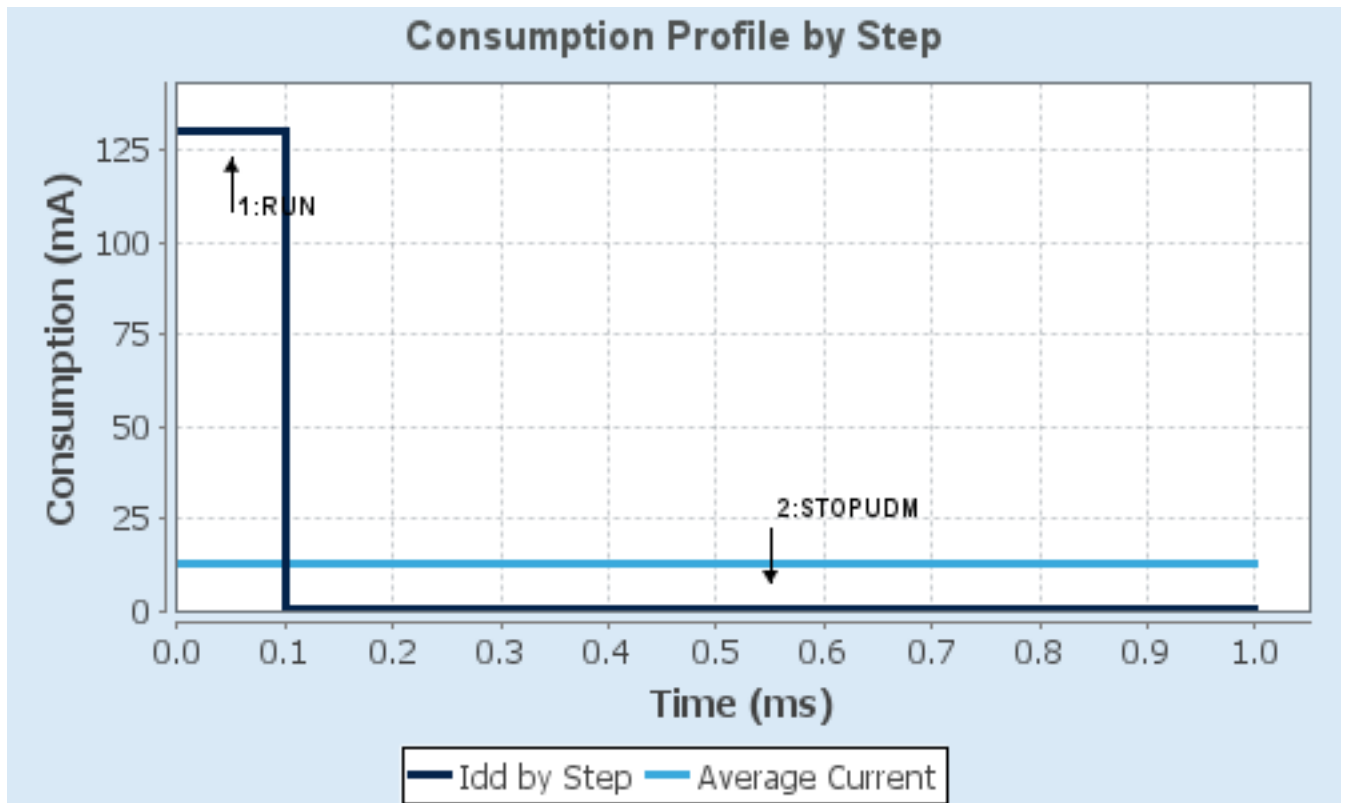
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP_UDM (Under Drive)
Vdd	3.3	3.3
Voltage Source	Battery	Battery
Range	Scale1-High	No Scale
Fetch Type	ITCM/FLASH/REGON	n/a
CPU Frequency	216 MHz	0 Hz
Clock Configuration	HSE PLL	Regulator_LP Flash-PwrDwn
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	130 mA	100 μ A
Duration	0.1 ms	0.9 ms
DMIPS	462.0	0.0
Ta Max	92.56	104.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	13.09 mA
Battery Life	1 day, 23 hours	Average DMIPS	462.24005 DMIPS

1.6. Chart



2. Peripherals and Middlewares Configuration

2.1. CORTEX_M7

2.1.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode Disabled

Cortex Interface Settings:

Flash Interface AXI Interface

ART ACCELERATOR Disabled

Instruction Prefetch Disabled

CPU ICache **Enabled ***

CPU DCache **Enabled ***

Cortex Memory Protection Unit Control Settings:

MPU Control Mode **Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers ***

Cortex Memory Protection Unit Region 0 Settings:

MPU Region **Enabled ***

MPU Region Base Address **0x00 ***

MPU Region Size **4GB ***

MPU SubRegion Disable **0x87 ***

MPU TEX field level **level 1 ***

MPU Access Permission ALL ACCESS NOT PERMITTED

MPU Instruction Access **DISABLE ***

MPU Shareability Permission **ENABLE ***

MPU Cacheable Permission DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region **Enabled ***

MPU Region Base Address **0x20048000 ***

MPU Region Size **16KB ***

MPU SubRegion Disable **0x0 ***

MPU TEX field level **level 1 ***

MPU Access Permission **ALL ACCESS PERMITTED ***

MPU Instruction Access ENABLE

MPU Shareability Permission DISABLE

MPU Cacheable Permission DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 2 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x2004C000 *
MPU Region Size	1KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	ENABLE
MPU Shareability Permission	ENABLE *
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled
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2.2. ETH

Mode: RMII

2.2.1. Parameter Settings:

General : Ethernet Configuration:

Warning	The ETH can work only when RAM is pointing at 0x24000000
Note	PHY Driver must be configured from the LwIP 'Platform Settings' top right tab
Ethernet MAC Address	00:80:E1:00:00:00
Tx Descriptor Length	4
First Tx Descriptor Address	0x2004C0a0 *
Rx Descriptor Length	4
First Rx Descriptor Address	0x2004C000 *
Rx Buffers Length	1536
Rx Mode	Interrupt Mode

2.3. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

2.3.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	6 WS (7 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Over Drive	Enabled
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1

2.4. SYS

Debug: Trace Asynchronous Sw

Timebase Source: TIM2

2.5. FREERTOS

Interface: CMSIS_V2

2.5.1. Config parameters:

API:

FreeRTOS API	CMSIS v2
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Versions:

FreeRTOS version	10.2.1
CMSIS-RTOS version	2.00

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock

TICK_RATE_HZ	1000
MAX_PRIORITIES	56
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	64 *
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	25*1024 *
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Option2 *

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Enabled *

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

2.5.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Disabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

2.5.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Enabled *
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Project settings (see parameter description first):

Use FW pack heap file	Enabled
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2.6. LWIP

mode: Enabled

Advanced parameters are not listed except if modified by user.

2.6.1. General Settings:

LwIP Version:

LWIP Version (Version of LwIP supported by CubeMX ** CubeMX specific **) 2.1.2

IPv4 - DHCP Options:

LWIP_DHCP (DHCP Module) **Disabled ***

IP Address Settings:

IP_ADDRESS (IP Address) **192.168.000.111 ***

NETMASK_ADDRESS (Netmask Address) **255.255.255.000 ***

GATEWAY_ADDRESS (Gateway Address) **192.168.000.001 ***

RTOS Dependency:

WITH_RTOS (Use FREERTOS ** CubeMX specific **) Enabled

CMSIS_VERSION (CMSIS API Version used) CMSIS v2

RTOS_USE_NEWLIB_REENTRANT (RTOS used - 1) Enabled

MBEDTLS Dependency:

WITH_MBEDTLS (Use MBEDTLS ** CubeMX specific **) Enabled

Platform Settings:

PHY Driver Choose/LAN8742/DP83848

Protocols Options:

LWIP_ICMP (ICMP Module Activation) Enabled

LWIP_IGMP (IGMP Module) Disabled

LWIP_DNS (DNS Module) Enabled

LWIP_UDP (UDP Module) Enabled

MEMP_NUM_UDP_PCB (Number of UDP Connections) 4

LWIP_TCP (TCP Module) Enabled

MEMP_NUM_TCP_PCB (Number of TCP Connections) 5

2.6.2. Key Options:

Infrastructure - OS Awareness Option:

NO_SYS (OS Awareness) OS Used

Infrastructure - Timers Options:

LWIP_TIMERS (Use Support For sys_timeout) Enabled

Infrastructure - Core Locking and MPU Options:

SYS_LIGHTWEIGHT_PROT (Memory Functions Protection) Enabled

Infrastructure - Heap and Memory Pools Options:

MEM_SIZE (Heap Memory Size) **16*1048 ***

Infrastructure - Internal Memory Pool Sizes:

MEMP_NUM_PBUF (Number of Memory Pool struct Pbufs) 16

MEMP_NUM_RAW_PCB (Number of Raw Protocol Control Blocks)	4
MEMP_NUM_TCP_PCB_LISTEN (Number of Listening TCP Connections)	16 *
MEMP_NUM_TCP_SEG (Number of TCP Segments simultaneously queued)	24 *
MEMP_NUM_LOCALHOSTLIST (Number of Host Entries in the Local Host List)	1

Pbuf Options:

PBUF_POOL_SIZE (Number of Buffers in the Pbuf Pool)	16
PBUF_POOL_BUFSIZE (Size of each pbuf in the pbuf pool)	592

IPv4 - ARP Options:

LWIP_ARP (ARP Functionality)	Enabled
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Callback - TCP Options:

TCP_TTL (Number of Time-To-Live Used by TCP Packets)	255
TCP_WND (TCP Receive Window Maximum Size)	8192 *
TCP_QUEUE_OOSEQ (Allow Out-Of-Order Incoming Packets)	Enabled
LWIP_TCP_SACK_OUT (Allow Sending Selective Acknowledgements)	Disabled
TCP_MSS (Maximum Segment Size)	1024 *
TCP_SND_BUF (TCP Sender Buffer Space)	2048
TCP_SND_QUEUELEN (Number of Packet Buffers Allowed for TCP Sender)	9

Network Interfaces Options:

LWIP_NETIF_STATUS_CALLBACK (Callback Function on Interface Status Changes)	Disabled
LWIP_NETIF_EXT_STATUS_CALLBACK (Extended Callback Function for several netif)	Disabled
LWIP_NETIF_LINK_CALLBACK (Callback Function on Interface Link Changes)	Enabled

NETIF - Loopback Interface Options:

LWIP_NETIF_LOOPBACK (NETIF Loopback)	Disabled
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Infrastructure - Threading Options:

TCPIP_THREAD_NAME (TCPIP Thread Name)	"tcpip_thread"
TCPIP_THREAD_STACKSIZE (TCPIP Thread Stack Size)	5*1024 *
TCPIP_THREAD_PRIO (TCPIP Thread Priority Level)	24
TCPIP_MBOX_SIZE (TCPIP Mailbox Size)	6
DEFAULT_THREAD_NAME (Default LwIP Thread Name)	"lwip"
DEFAULT_THREAD_STACKSIZE (Default LwIP Thread Stack Size)	1024
DEFAULT_THREAD_PRIO (Default LwIP Thread Priority Level)	3
DEFAULT_RAW_RECVMBOX_SIZE (Default Mailbox Size on a NETCONN Raw)	0
DEFAULT_TCP_RECVMBOX_SIZE (Default Mailbox Size on a NETCONN TCP)	6
DEFAULT_ACCEPTMBOX_SIZE (Default Mailbox Size for Incoming Connections)	6

Thread Safe APIs - Netconn Options:

LWIP_NETCONN (NETCONN API)	Enabled
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Thread Safe APIs - Socket Options:

LWIP_SOCKET (Socket API)	Enabled
LWIP_COMPAT_SOCKETS (BSD-style Socket Functions Names)	1
LWIP_SOCKET_OFFSET (Socket Offset Number)	0

LWIP_TCP_KEEPALIVE (TCP_KEEPIIDLE, TCP_KEEPINTVL and TCP_KEEPCNT Options)	Enabled *
LWIP_SOCKET_SELECT (Select for Socket)	Enabled
LWIP_SOCKET_POLL (Poll for Socket)	Enabled

2.6.3. PPP:

PPP Options:

PPP_SUPPORT (PPP Module)	Disabled
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2.6.4. IPv6:

IPv6 Options:

LWIP_IPV6 (IPv6 Protocol)	Disabled
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2.6.5. HTTPD:

HTTPD Options:

LWIP_HTTPD (LwIP HTTPD Support ** CubeMX specific **)	Enabled *
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2.6.6. SNMP:

SNMP Options:

LWIP_SNMP (LwIP SNMP Agent)	Disabled
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2.6.7. SNTP/SMTP:

SNTP Options:

LWIP_SNTP (LWIP SNTP Support ** CubeMX specific **)	Disabled
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SMTP Options:

LWIP_SMTP (LWIP SMTP Support ** CubeMX specific **)	Disabled
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2.6.8. MDNS/TFTP:

MDNS Options:

LWIP_MDNS (Multicast DNS Support ** CubeMX specific **)	Disabled
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TFTP Options:

LWIP_TFTP (TFTP Support ** CubeMX specific **)	Disabled
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2.6.9. Perf/Checks:

Sanity Checks:

LWIP_DISABLE_TCP_SANITY_CHECKS (TCP Sanity Checks)	Disabled
LWIP_DISABLE_MEMP_SANITY_CHECKS (MEMP Sanity Checks)	Disabled

Performance Options:

LWIP_PERF (Performace Testing for LwIP)	Disabled
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2.6.10. Statistics:

Debug - Statistics Options:

LWIP_STATS (Statistics Collection)	Disabled
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2.6.11. Checksum:

Infrastructure - Checksum Options:

CHECKSUM_BY_HARDWARE (Hardware Checksum ** CubeMX specific **)	Enabled
LWIP_CHECKSUM_CTRL_PER_NETIF (Generate/Check Checksum per Netif)	Disabled
CHECKSUM_GEN_IP (Generate Software Checksum for Outgoing IP Packets)	Disabled
CHECKSUM_GEN_UDP (Generate Software Checksum for Outgoing UDP Packets)	Disabled
CHECKSUM_GEN_TCP (Generate Software Checksum for Outgoing TCP Packets)	Enabled *
CHECKSUM_GEN_ICMP (Generate Software Checksum for Outgoing ICMP Packets)	Disabled
CHECKSUM_GEN_ICMP6 (Generate Software Checksum for Outgoing ICMP6 Packets)	Disabled
CHECKSUM_CHECK_IP (Generate Software Checksum for Incoming IP Packets)	Disabled
CHECKSUM_CHECK_UDP (Generate Software Checksum for Incoming UDP Packets)	Disabled
CHECKSUM_CHECK_TCP (Generate Software Checksum for Incoming TCP Packets)	Disabled
CHECKSUM_CHECK_ICMP (Generate Software Checksum for Incoming ICMP Packets)	Disabled
CHECKSUM_CHECK_ICMP6 (Generate Software Checksum for Incoming ICMP6 Packets)	Disabled

2.6.12. Debug:

LwIP Main Debugging Options:

LWIP_DBG_MIN_LEVEL (Minimum Level)	All
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2.6.13. Platform Settings:

Driver_PHY	LAN8742
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2.7. MBEDTLS

mode: Enabled

2.7.1. Version and modes:

Version:

MBEDTLS version 2.16.2

TCP/IP stack:

TCP/IP stack LWIP

RNG dependency:

RNG IP SW RNG(no entropy)

Modes:

MBEDTLS_SSL_CLI_C **Not Defined ***

MBEDTLS_SSL_SRV_C **Not Defined ***

2.7.2. Feature support:

System support:

MBEDTLS_HAVE_ASM **Not Defined ***

MBEDTLS_NO_UDBL_DIVISION **Not Defined ***

MBEDTLS_HAVE_TIME **Not Defined ***

MBEDTLS_HAVE_TIME_DATE **Not Defined ***

General:

MBEDTLS_ECP_NIST_OPTIM **Not Defined ***

MBEDTLS_ECDSA_DETERMINISTIC Not Defined

MBEDTLS_PK_PARSE_EC_EXTENDED **Not Defined ***

MBEDTLS_ERROR_STRERROR_DUMMY **Not Defined ***

MBEDTLS_GENPRIME Not Defined

MBEDTLS_NO_DEFAULT_ENTROPY_SOURCES **Not Defined ***

MBEDTLS_NO_PLATFORM_ENTROPY **Not Defined ***

MBEDTLS_PK_RSA_ALT_SUPPORT **Not Defined ***

MBEDTLS_PKCS1_V15 Not Defined

MBEDTLS_PKCS1_V21 Not Defined

MBEDTLS_SELF_TEST **Not Defined ***

MBEDTLS_VERSION_FEATURES Not Defined

Ciphering:

MBEDTLS_CIPHER_MODE_CBC	Not Defined *
MBEDTLS_CIPHER_MODE_CFB	Not Defined *
MBEDTLS_CIPHER_MODE_CTR	Not Defined *
MBEDTLS_CIPHER_MODE_OFB	Not Defined *
MBEDTLS_CIPHER_MODE_XTS	Not Defined *
MBEDTLS_CIPHER_PADDING_PKCS7	Not Defined *
MBEDTLS_CIPHER_PADDING_ONE_AND_ZEROS	Not Defined *
MBEDTLS_CIPHER_PADDING_ZEROS_AND_LEN	Not Defined *
MBEDTLS_CIPHER_PADDING_ZEROS	Not Defined *
MBEDTLS_REMOVE_ARC4_CIPHERSUITES	Not Defined *
MBEDTLS_REMOVE_3DES_CIPHERSUITES	Not Defined *

Elliptic curves:

MBEDTLS_ECP_DP_SECP192R1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_SECP224R1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_SECP256R1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_SECP384R1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_SECP521R1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_SECP192K1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_SECP224K1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_SECP256K1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_BP256R1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_BP384R1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_BP512R1_ENABLED	Not Defined *
MBEDTLS_ECP_DP_CURVE25519_ENABLED	Not Defined *
MBEDTLS_ECP_DP_CURVE448_ENABLED	Not Defined *

Key exchange:

MBEDTLS_KEY_EXCHANGE_PSK_ENABLED	Not Defined *
MBEDTLS_KEY_EXCHANGE_DHE_PSK_ENABLED	Not Defined
MBEDTLS_KEY_EXCHANGE_ECDHE_PSK_ENABLED	Not Defined
MBEDTLS_KEY_EXCHANGE_RSA_PSK_ENABLED	Not Defined
MBEDTLS_KEY_EXCHANGE_RSA_ENABLED	Not Defined
MBEDTLS_KEY_EXCHANGE_DHE_RSA_ENABLED	Not Defined
MBEDTLS_KEY_EXCHANGE_ECDHE_RSA_ENABLED	Not Defined
MBEDTLS_KEY_EXCHANGE_ECDH_ECDSA_ENABLED	Not Defined
MBEDTLS_KEY_EXCHANGE_ECDH_RSA_ENABLED	Not Defined

SSL:

MBEDTLS_SSL_ALL_ALERT_MESSAGES	Not Defined
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MBEDTLS_SSL_ENCRYPT_THEN_MAC	Not Defined
MBEDTLS_SSL_EXTENDED_MASTER_SECRET	Not Defined
MBEDTLS_SSL_FALLBACK_SCSV	Not Defined
MBEDTLS_SSL_RENEGOTIATION	Not Defined
MBEDTLS_SSL_PROTO_TLS1	Not Defined
MBEDTLS_SSL_PROTO_TLS1_1	Not Defined
MBEDTLS_SSL_PROTO_DTLS	Not Defined
MBEDTLS_SSL_DTLSANTI_REPLAY	Not Defined
MBEDTLS_SSL_DTLS_HELLO_VERIFY	Not Defined
MBEDTLS_SSL_DTLS_CLIENT_PORT_REUSE	Not Defined
MBEDTLS_SSL_DTLS_BADMAC_LIMIT	Not Defined
MBEDTLS_SSL_SERVER_NAME_INDICATION	Not Defined

X509:

MBEDTLS_X509_CHECK_KEY_USAGE	Not Defined *
MBEDTLS_X509_CHECK_EXTENDED_KEY_USAGE	Not Defined *
MBEDTLS_X509_RSASSA_PSS_SUPPORT	Not Defined *

2.7.3. Alternate implementation:

2.7.4. Modules:

General:

MBEDTLS_AESNI_C	Not Defined
MBEDTLS_AES_C	Not Defined *
MBEDTLS_ARC4_C	Not Defined *
MBEDTLS_ASN1_PARSE_C	Not Defined *
MBEDTLS_ASN1_WRITE_C	Not Defined *
MBEDTLS_BASE64_C	Defined
MBEDTLS_BIGNUM_C	Not Defined *
MBEDTLS_BLOWFISH_C	Not Defined *
MBEDTLS_CAMELLIA_C	Not Defined *
MBEDTLS_CCM_C	Not Defined
MBEDTLS_CERTS_C	Not Defined *
MBEDTLS_CIPHER_C	Not Defined *
MBEDTLS_CHACHA20_C	Not Defined *
MBEDTLS_CHACHAPOLY_C	Not Defined
MBEDTLS_CTR_DRBG_C	Not Defined
MBEDTLS_DES_C	

MBEDTLS_DHM_C	Not Defined *
MBEDTLS_ECDH_C	Not Defined *
MBEDTLS_ECDSA_C	Not Defined
MBEDTLS_ECP_C	Not Defined
MBEDTLS_ENTROPY_C	Not Defined
MBEDTLS_ERROR_C	Not Defined *
MBEDTLS_GCM_C	Not Defined
MBEDTLS_HKDF_C	Not Defined *
MBEDTLS_HMAC_DRBG_C	Not Defined
MBEDTLS_MD_C	Not Defined *
MBEDTLS_MD5_C	Not Defined *
MBEDTLS_NET_C	Not Defined *
MBEDTLS_NIST_KW_C	Not Defined
MBEDTLS_OID_C	Not Defined *
MBEDTLS_PADLOCK_C	Not Defined
MBEDTLS_PEM_PARSE_C	Defined
MBEDTLS_PEM_WRITE_C	Defined
MBEDTLS_PK_C	Not Defined
MBEDTLS_PK_PARSE_C	Not Defined
MBEDTLS_PK_WRITE_C	Not Defined
MBEDTLS_PKCS5_C	Not Defined
MBEDTLS_PKCS12_C	Not Defined
MBEDTLS_PLATFORM_C	Not Defined *
MBEDTLS_RIPEMD160_C	Not Defined *
MBEDTLS_POLY1305_C	Not Defined *
MBEDTLS_RSA_C	Not Defined
MBEDTLS_SHA1_C	Defined
MBEDTLS_SHA256_C	Not Defined *
MBEDTLS_SHA512_C	Not Defined *
MBEDTLS_SSL_TICKET_C	Not Defined
MBEDTLS_SSL_TLS_C	Not Defined
MBEDTLS_VERSION_C	Not Defined *
MBEDTLS_X509_USE_C	Not Defined
MBEDTLS_X509_CRT_PARSE_C	Not Defined
MBEDTLS_X509_CRL_PARSE_C	Not Defined
MBEDTLS_X509_CSR_PARSE_C	Not Defined
MBEDTLS_X509_CREATE_C	Not Defined
MBEDTLS_X509_CRT_WRITE_C	Not Defined
MBEDTLS_X509_CSR_WRITE_C	Not Defined

MBEDTLS_XTEA_C

Not Defined *

2.7.5. Modules Configuration:

*** User modified value**

3. System Configuration

3.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ETH	PG14	ETH_TXD1	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PG13	ETH_TXD0	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PG11	ETH_TX_EN	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC1	ETH_MDC	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA1	ETH_REF_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC4	ETH_RXD0	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA2	ETH_MDIO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PC5	ETH_RXD1	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA7	ETH_CRS_DV	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
RCC	PH0/OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1/OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SYS	PB3	SYS_JTDO-SWO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
GPIO	PI1	GPIO_Output	Output Push Pull	Pull-down *	Low	LD1
	PI11	GPIO_Input	Input mode	Pull-down *	n/a	B1_USER

3.2. DMA configuration

nothing configured in DMA service

3.3. NVIC configuration

3.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
TIM2 global interrupt	true	15	0
Ethernet global interrupt	true	15	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
Ethernet wake-up interrupt through EXTI line 19	unused		
FPU global interrupt	unused		

3.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
TIM2 global interrupt	false	true	true
Ethernet global interrupt	false	true	true

* User modified value

4. System Views

4.1. Category view

4.1.1. Current

Middlewares

FREERTOS ✓

LWIP ✓

MBEDTLS ✓

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing
CORTEX_M7 ✓ DMA GPIO ✓ IVIC ✓ RCC ✓ SYS ✓			ETH ✓			

5. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32f7_bsd.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32f7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32f7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
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Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstmcsuite.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32gui.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-stm32-microcontroller-system-memory-boot-mode-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack-mcus-and-mpus-stmicroelectronics.pdf

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- Application Notes https://www.st.com/resource/en/application_note/an4031-using-the-stm32f2-stm32f4-and-stm32f7-series-dma-controller-stmicroelectronics.pdf
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- Application Notes https://www.st.com/resource/en/application_note/an4229-how-to-implement-a-vocoder-solution-using-stm32-microcontrollers-stmicroelectronics.pdf
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- Application Notes https://www.st.com/resource/en/application_note/an4731-stm32cube-mcu-package-examples-for-stm32f7-series-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an4946-migration-of-microcontroller-applications-between-stm32f72xxx73xxx-and-stm32f74xxx75xxx-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf

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of-stm32-mcus-and-mpus-stmicroelectronics.pdf

- Application Notes https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
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& Software

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& Software

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for related Tools & Software	graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
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Errata Sheets	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/errata_sheet/es0290-stm32f74xxx-and-stm32f75xxx-device-limitations-stmicroelectronics.pdf
Programming Manuals	https://www.st.com/resource/en/datasheet/dm00166116.pdf
Reference Manuals	https://www.st.com/resource/en/programming_manual/pm0253-stm32f7-series-and-stm32h7-series-cortexm7-processor-programming-manual-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/reference_manual/rm0385-stm32f75xxx-and-stm32f74xxx-advanced-armbased-32bit-mcus-stmicroelectronics.pdf

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User Manuals	https://www.st.com/resource/en/user_manual/um2318-stm32f7-series-safety-manual-stmicroelectronics.pdf
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